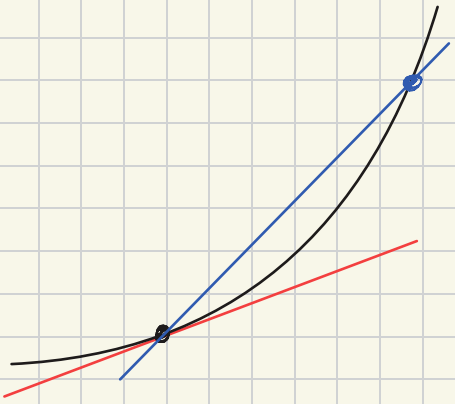
§2.2

The definition of Derivative

(First Principle of Derivative)

$$f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h} = \lim_{x\to a} \frac{f(x)-f(a)}{x-a}$$



If $f'(a)$ exists, we say f is differentiable at $x=a$.

The Derivation of Function: $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$

$f(x) = |x|$

$f'(x) = \begin{cases} 1 & \text{if } x>0 \\ -1 & \text{if } x<0 \end{cases}$

Notation:

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx}f(x) = D_x[y]$$

$\frac{d}{dx}$: differential operator, not d divided by dx.

Thm 2.1.

f is differentiable $\Rightarrow$ f is continuous.

$\not\Leftarrow$: e.g. $|x|$

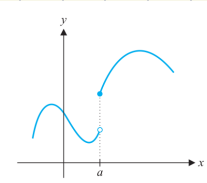


FIGURE 2.19a
A jump discontinuity

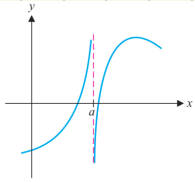


FIGURE 2.19b
A vertical asymptote

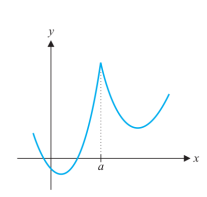


FIGURE 2.19c
A cusp

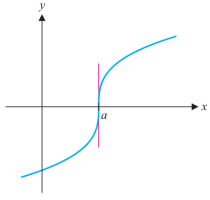


FIGURE 2.19d
A vertical tangent line

Appendix: L-R derivative

1. Left Derivative: $f'_-(x) = \lim_{h\to 0^-} \frac{f(x+h)-f(x)}{h}$
2. Right Derivative: $f'_+(x) = \lim_{h\to 0^+} \frac{f(x+h)-f(x)}{h}$
3. f is diff $\Leftrightarrow$ $f'_-(x) = f'_+(x) = f'(x)$

In exercises 1–4, compute $f'(a)$ using the limits (2.1) and (2.2).

$\lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$ ✓

$\lim_{x\to a} \frac{f(x)-f(a)}{x-a}$

4. $f(x)=\frac{3}{x+1}$, $a=2$, $f(2)=1$

$$f'(2)=\lim_{h\to 0}\frac{f(2+h)-f(2)}{h}=\lim_{h\to 0}\frac{\frac{3}{3+h}-1}{h}$$

$$=\lim_{h\to 0}\frac{\frac{3-3-h}{3+h}}{h}=\lim_{h\to 0}\frac{-h}{h(3+h)}$$

$$=\frac{-1}{3}\ \#$$

In exercises 5–12, compute the derivative function f' using (2.1) or (2.2).

8. $f(x)=x^4-2x^2+1=(x^2-1)^2$

$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$

$$f'(x)=\lim_{h\to 0}\frac{((x+h)^2-1)^2-(x^2-1)^2}{h}$$

$$=\lim_{h\to 0}\frac{((x+h)^2-1+x^2-1)((x+h)^2-1-x^2+1)}{h}$$

$(x+h)^2-x^2=(2x+h)\cdot h$

$$=\lim_{h\to 0}\frac{((x+h)^2-1+x^2-1)(2x+h)\cdot h}{h}$$

$$=(x^2-1+x^2-1)\cdot 2x=(2x^2-2)\cdot 2x=4x^3-4x\ \#$$

11. $f(t)=\sqrt{3t+1}$

$$f'(t)=\lim_{h\to 0}\frac{\sqrt{3(t+h)+1}-\sqrt{3t+1}}{h}\cdot\frac{\sqrt{3(t+h)+1}+\sqrt{3t+1}}{\sqrt{3(t+h)+1}+\sqrt{3t+1}}$$

$$=\lim_{h\to 0}\frac{3(t+h)+1-3t-1}{h(\sqrt{3(t+h)+1}+\sqrt{3t+1})}=\lim_{h\to 0}\frac{3t+3h-3t}{h(\sqrt{3(t+h)+1}+\sqrt{3t+1})}=\lim_{h\to 0}\frac{3h}{h(\sqrt{3(t+h)+1}+\sqrt{3t+1})}$$

$$=\frac{3}{\sqrt{3t+1}+\sqrt{3t+1}}=\frac{3}{2\sqrt{3t+1}}\ \#$$

In exercises 19–22, compute the right-hand derivative $D_+ f(0) = \lim_{h \to 0^+} \dfrac{f(h) - f(0)}{h}$ and the left-hand derivative $D_- f(0) = \lim_{h \to 0^-} \dfrac{f(h) - f(0)}{h}$. Does $f'(0)$ exist?

22. $f(x) = \begin{cases} 2x & \text{if } x < 0 \\ x^2 + 2x & \text{if } x \geq 0 \end{cases}$ ✓

$f(0) = 0$

$f'_-(0) = \lim_{h \to 0^-} \dfrac{f(0+h) - f(0)}{h} = \lim_{h \to 0^-} \dfrac{2h}{h} = 2$

$f'_+(0) = \lim_{h \to 0^+} \dfrac{f(0+h) - f(0)}{h} = \lim_{h \to 0^+} \dfrac{h^2 + 2h}{h} = 2$

Since $f'_-(0) = f'_+(0) = 2$, $f'(0) = 2$. #

28. For $f(x) = \begin{cases} x^2 + 2x, & \text{if } x \leq 0 \\ ax + b, & \text{if } x > 0 \end{cases}$ find all numbers a and b such that $f'(0)$ exists.

$f(0) = 0$

$f'_-(0) = \lim_{h \to 0^-} \dfrac{h^2 + 2h}{h} = 2$

$f'_+(0) = \lim_{h \to 0^+} \dfrac{ah + b}{h} = \lim_{h \to 0^+} a + \dfrac{b}{h}$

$= \lim_{h \to 0^+} a = a$

$\lim_{x \to 0^+} ax + b = 0 \Rightarrow b = 0$

Hence $a = 2$, $b = 0$ #

31. If f is differentiable at $x = a \neq 0$, ~~evaluate~~

$$\lim_{x \to a} \frac{[f(x)]^2 - [f(a)]^2}{x^2 - a^2}.$$

$$\lim_{x \to a} \frac{[f(x)]^2 - [f(a)]^2}{x^2 - a^2} = \lim_{x \to a} \frac{(f(x) + f(a))(f(x) - f(a))}{(x + a)(x - a)}$$

$$= \left(\lim_{x \to a} \frac{f(x) + f(a)}{x + a} \right) \cdot \left(\lim_{x \to a} \frac{f(x) - f(a)}{x - a} \right)$$

$$= \frac{f(a)}{a} \cdot f'(a) \quad \text{※}$$

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

§ 2.3

Power Rule: $\frac{d}{dx}(x^n) = n \cdot x^{n-1}$, $n \in \mathbb{N}$.

(① $\frac{d}{dx} C = 0$, $C \in \mathbb{R}$ ② $\frac{d}{dx} x = 1$)

$a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + a^{n-3}b^2 + \cdots + ab^{n-2} + b^{n-1})$

pf Let $f(x) = x^n$, then

$$f'(x) = \lim_{h\to 0} \frac{(x+h)^n - x^n}{h}$$

$$= \lim_{h\to 0} \frac{(x+h-x)\left[(x+h)^{n-1} + (x+h)^{n-2}\cdot x + \cdots + x^{n-1}\right]}{h}$$

$$= \lim_{h\to 0} \frac{h\left[(x+h)^{n-1} + (x+h)^{n-2}\cdot x + \cdots + x^{n-1}\right]}{h}$$

$$= \lim_{h\to 0} (x+h)^{n-1} + (x+h)^{n-2}\cdot x + \cdots + x^{n-1}$$

$$= \underbrace{x^{n-1} + x^{n-1} + \cdots + x^{n-1}}_{n\text{個}} = n \cdot x^{n-1}. \quad //$$

General Power Rule: $\forall r \neq 0$, $\frac{d}{dx} x^r = r x^{r-1}$.

General Derivative Rule: If f, g are differentiable at x, $C \in \mathbb{R}$.

1. $(f \pm g)'(x) = f'(x) \pm g'(x)$
2. $(cf)'(x) = c f'(x)$
3. $(fg)'(x) = f'(x)g(x) + f(x)g'(x)$
4. If $g(x) \neq 0$, then $\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$.

} §2.4

5. General Product Rule

For $n \in \mathbb{N}$, $\left(\prod_{k=1}^{n} f_k(x)\right)' = ?$

Higher Order Derivative

$(y')' = y''$

First-Order Derivative: $f' = \frac{dy}{dx} = y'$

Second-Order Derivative: $(f')' = f'' = \frac{d^2y}{dx^2} = y'' = \frac{d}{dx}\left(\frac{dy}{dx}\right)$

Third-Order Derivative: $(f'')' = f''' = \frac{d^3y}{dx^3} = y'''$

4th-Order Derivative: $f^{(4)} = \frac{d^4y}{dx^4} = y^{(4)}$ $\quad f^{(4)}(x)\ \frac{d^4}{dx^4}y$

$\vdots$

nth-Order Derivative: $f^{(n)} = \frac{d^ny}{dx^n} = y^{(n)}$ $\quad f^{(n)}(x) = \frac{d^n}{dx^n}y$

In exercises 1–14, differentiate each function.

8. $h(x) = 12x - x^2 - \frac{2}{\sqrt[3]{x^2}} = 12x - x^2 - 2x^{-\frac{2}{3}}$ $\quad 2x^{\frac{1}{3}}$

$h'(x) = 12 - 2x + \frac{4}{3}x^{-\frac{5}{3}}$

12. $f(x) = \frac{4x^2 - x - 3}{\sqrt{x}} = 4x^{\frac{3}{2}} - x^{\frac{1}{2}} - 3x^{-\frac{1}{2}}$ $\quad \sqrt{x} = x^{\frac{1}{2}}$

$f'(x) = 6x^{\frac{1}{2}} - \frac{1}{2}x^{-\frac{1}{2}} + \frac{3}{2}x^{-\frac{3}{2}}$

In exercises 15–20, compute the indicated derivative.

17. $\frac{d^2f}{dx^2}$ for $f(x) = 2x^4 - \frac{3}{\sqrt{x}} = 2x^4 - 3x^{-\frac{1}{2}}$

$\frac{df}{dx} = 8x^3 + \frac{3}{2}x^{-\frac{3}{2}}$

$\frac{d^2f}{dx^2} = 24x^2 - \frac{9}{4}x^{-\frac{5}{2}}$ ✱

19. $f^{(4)}(x)$ for $f(x) = x^4 + 3x^2 - \frac{2}{\sqrt{x}} = x^4 + 3x^2 - 2x^{\frac{-1}{2}}$

$f'(x) = 4x^3 + 6x + x^{\frac{-3}{2}}$

$f''(x) = 12x^2 + 6 - \frac{3}{2}x^{\frac{-5}{2}}$

$f'''(x) = 24x + \frac{15}{4}x^{\frac{-7}{2}}$

$f^{(4)}(x) = 24 - \frac{105}{8}x^{\frac{-9}{2}}$ #

In exercises 21–24, use the given position function to find the velocity and acceleration functions.

23. $S(t) = \sqrt{t} + 2t^2$ $\quad (\sqrt{x})' = \frac{1}{2\sqrt{x}} = \frac{1}{2}x^{\frac{-1}{2}}$

$V(t) = S'(t) = \frac{1}{2\sqrt{t}} + 4t$

$a(t) = V'(t) = \frac{-1}{4}x^{\frac{-3}{2}} + 4$

In exercises 27–30, find an equation of the tangent line to $y = f(x)$ at $x = a$.

27. $f(x) = x^2 - 2, a = 2$ **28.** $f(x) = x^2 - 2x + 1, a = 2$

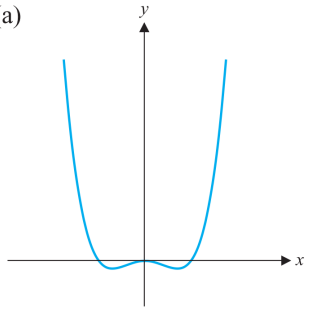
29. $f(x) = 4\sqrt{x} - 2x, a = 4$ **30.** $f(x) = 3\sqrt{x} + 4, a = 2$

$f'(x) = \frac{3}{2\sqrt{x}}$, $f'(2) = \frac{3}{2\sqrt{2}}$, $f(2) = 3\sqrt{2} + 4$

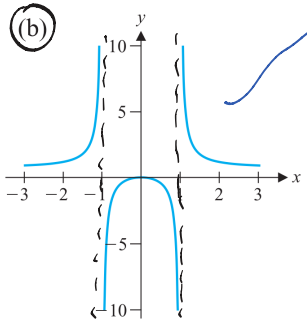
$y - (3\sqrt{2} + 4) = \frac{3}{2\sqrt{2}}(x - 2)$ #

In exercises 31 and 32, use the graph of f to sketch a graph of f''. (Hint: Sketch f' first.)

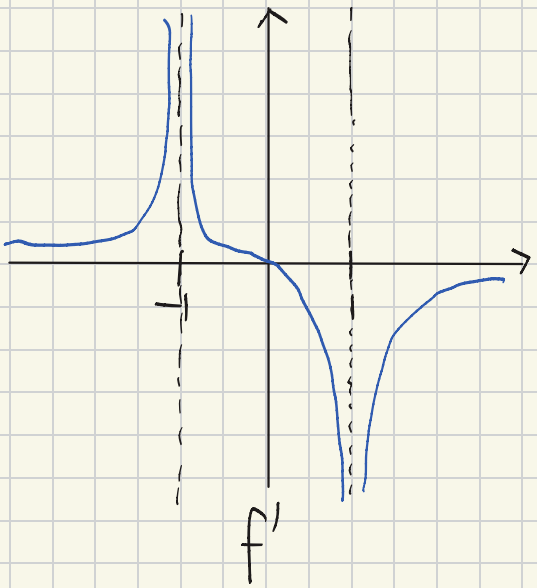
31. (a)

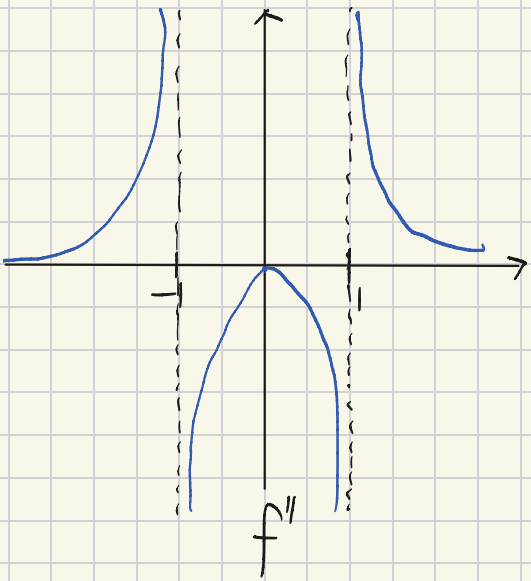


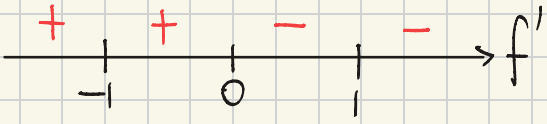
(b)

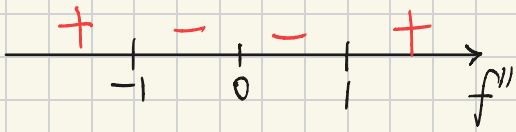


$$f(x) = \frac{x^2}{x^2-1}$$









39. Find a second-degree polynomial (of the form $f(x) = ax^2 + bx + c$) such that (a) $f(0) = -2$, $f'(0) = 2$ and $f''(0) = 3$. (b) $f(0) = 0$, $f'(0) = 5$ and $f''(0) = 1$.

(a) Given $f(x) = ax^2 + bx + c$

$f'(x) = 2ax + b$

$f''(x) = 2a$

Since $f(0) = c = -2$, $f'(0) = b = 2$, $f''(0) = 2a = 3$

$\Rightarrow a = \frac{3}{2}$, $b = 2$, $c = -2$

$\Rightarrow f(x) = \frac{3}{2}x^2 + 2x - 2$ ✱

40. Find a general formula for the nth derivative $f^{(n)}(x)$.

(a) $f(x) = \sqrt{x} = x^{\frac{1}{2}}$ (b) $f(x) = \frac{2}{x}$

$\frac{d^n}{dx^n}(x^n) = ?$

Ans: $n!$

(a) $f'(x) = \frac{1}{2} \cdot x^{\frac{-1}{2}}$

$f''(x) = \frac{1}{2} \cdot \left(\frac{-1}{2}\right) \cdot x^{\frac{-3}{2}}$

$f'''(x) = \frac{1}{2} \cdot \left(\frac{-1}{2}\right) \cdot \left(\frac{-3}{2}\right) \cdot x^{\frac{-5}{2}}$

$\vdots$

$f^{(n)}(x) = (-1)^{n+1} \cdot \frac{1 \cdot 3 \cdot 5 \cdots (2n-3)}{2^n} \cdot x^{\frac{-(2n-1)}{2}}$ ✱

In exercises 45–48, find a function with the given derivative.

45. $f'(x) = 4x^3$

46. $f'(x) = 5x^4$

47. $f'(x) = \sqrt{x} = x^{\frac{1}{2}}$

48. $f'(x) = \dfrac{1}{x^2}$

Hint: $(x^r)' = rx^{r-1}$

$r \in \mathbb{R} \setminus \{0\}$

$\Downarrow$

$\frac{1}{r}(x^r)' = x^{r-1}$

$=$

$(\frac{1}{r}x^r)'$

$=$

$f(x)$

47. Let $f(x) = \frac{1}{r}x^r$

$f'(x) = x^{r-1} = x^{\frac{1}{2}} \Rightarrow r-1 = \frac{1}{2}$

$\Rightarrow r = \frac{3}{2}$

Hence $f(x) = \frac{2}{3}x^{\frac{3}{2}}$. #